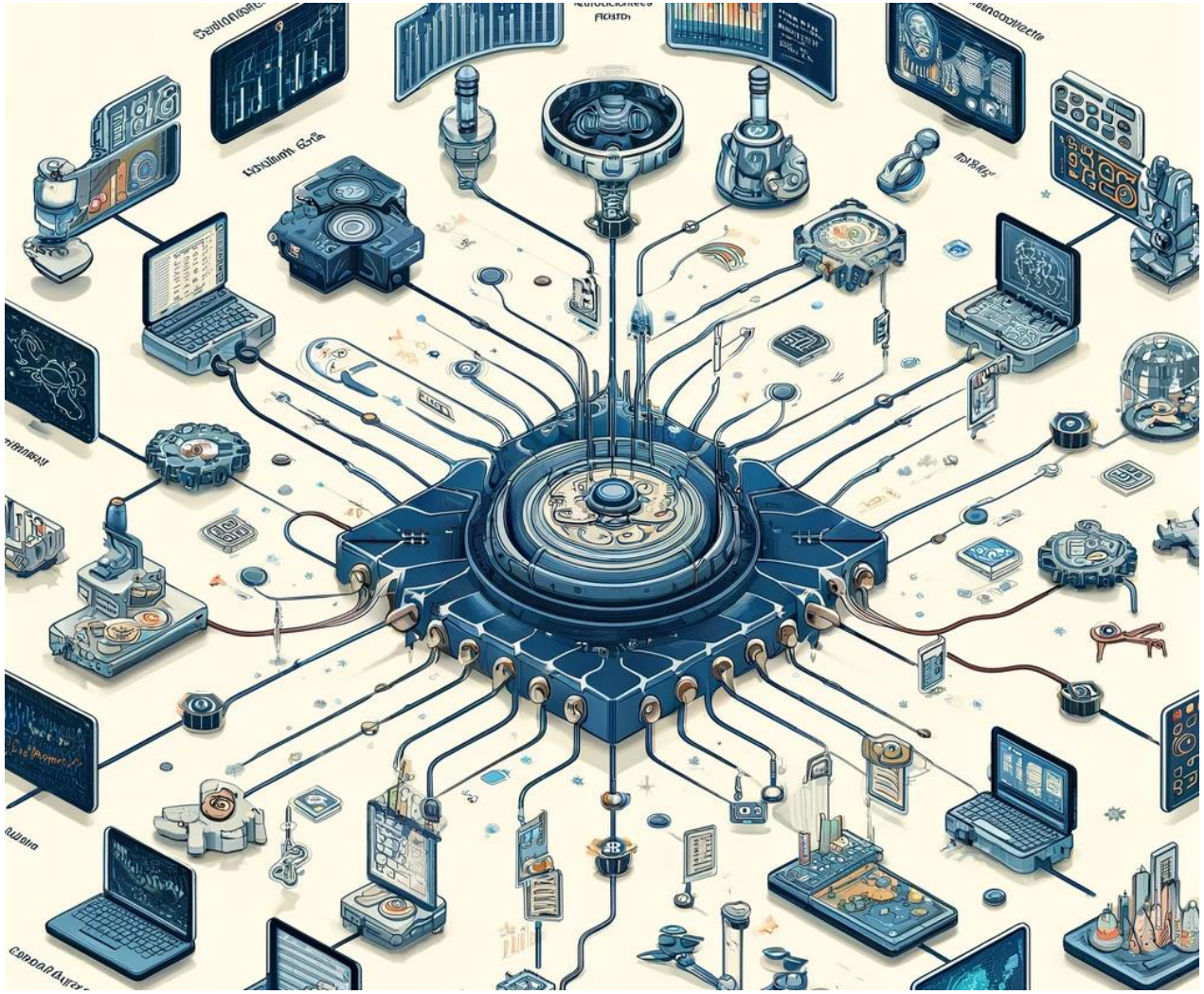
Exploring Scientific Workflows with CWL and dispel4py

Module 1.a

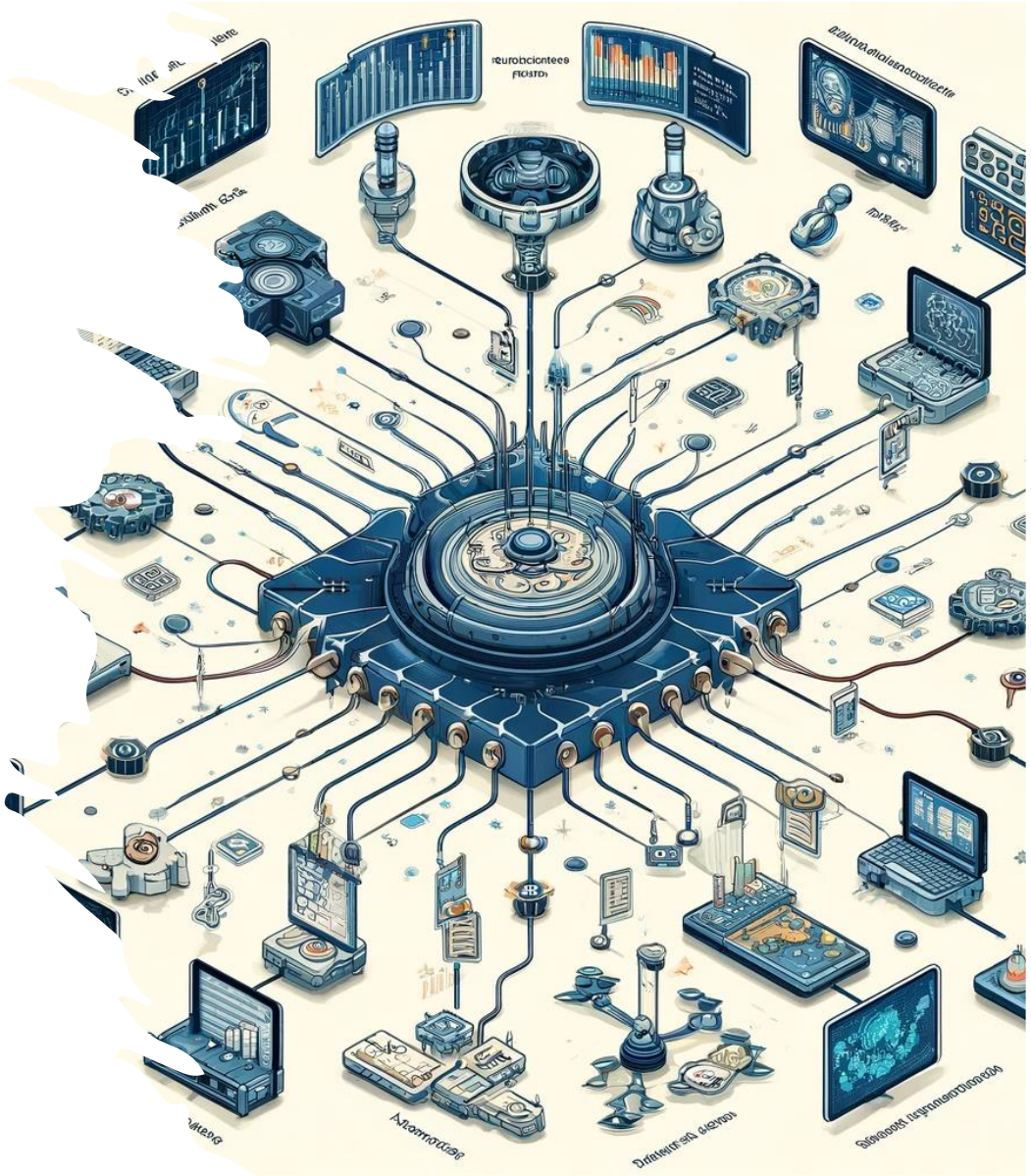
- Dr. Rosa Filgueira
- Chancellor’s Fellow AI & DS
- University of Edinburgh, EPCC
- r.filgueira@epcc.ed.ac.uk



Seminar Overview

- Day 1: Understanding Scientific Workflows (2 Hours)
  - Module 1: Introduction to Scientific Workflows
    - Defining scientific workflows and WMS
    - How to Select WMS
    - Workflows Concepts
  - Module 2: Creating Workflows with CWL

- Day 2: Exploring dispel4py and its applications (2 Hours)
  - Module 3: Introduction to dispel4py
  - Module 4: Hands-on Exercises



1. Defining scientific workflows & Workflow Management Systems (WMS)

Workflow Definition and Importance

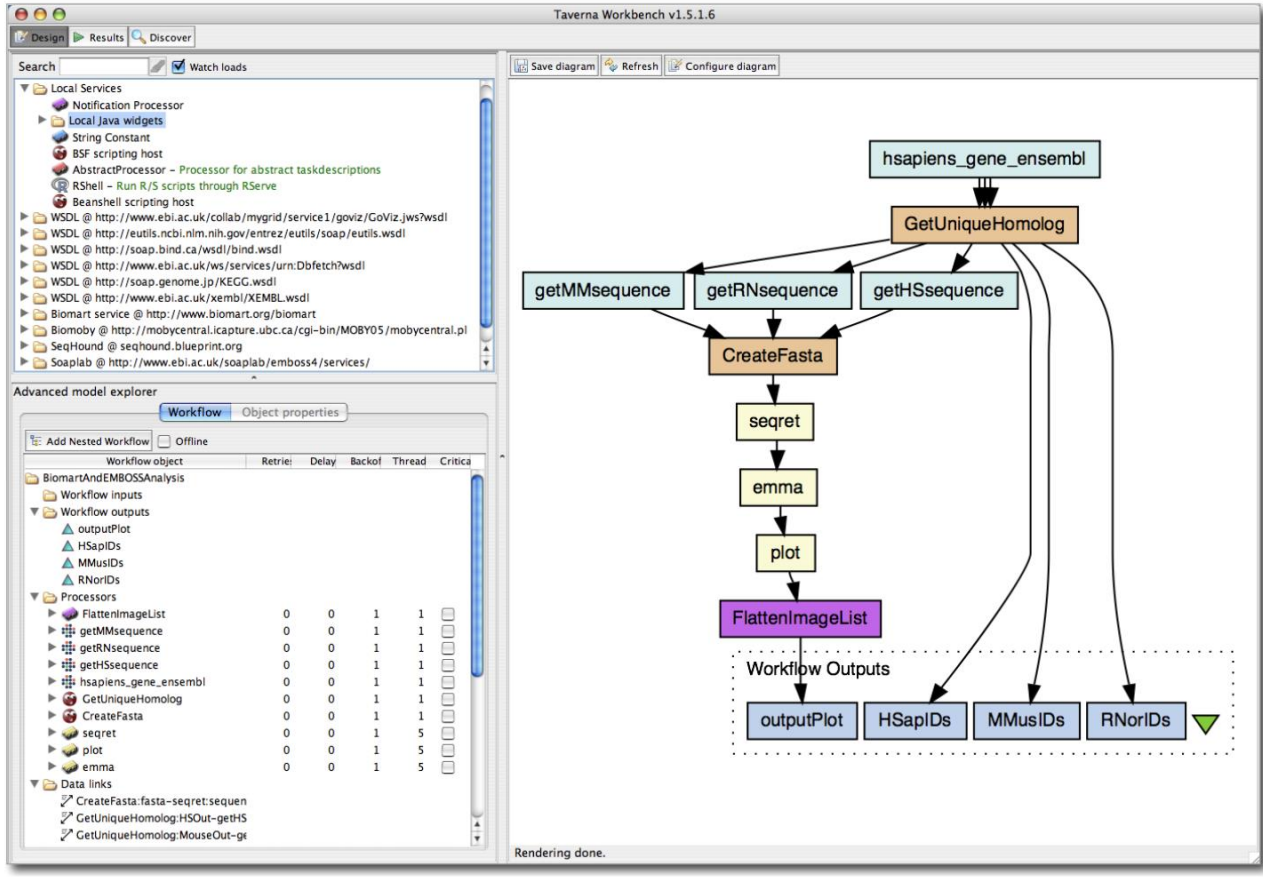
- Workflows refer to the structured sequences of:
  - Data processing
  - Analysis tasks
- Workflows are crucial for ensuring:
  - Systematically execution of complex and multi-step processes/tasks
    - Facilitate the organization of tasks
  - Verifiable and reproducible results.
    - Enhance the efficiency and reliability of scientific research.

Workflow Definition and Importance

- Formal way to express a calculation
- Multiple tasks with dependencies between them
- No limitations on tasks
  - Short or long
  - Loosely or tightly coupled
- Capture task parameters, input, output
- Independence of workflow process and data
- Often, run same workflow with different data

Workflow Definition and Importance

- Example workflow represented in the Taverna workflow system.
  - Extracts gene IDs from human chromosome 22 with mappings to disease functions and homologues in mouse and rat
  - Fetches base pairs of the associated DNA sequences;
  - Combines the sequences into a FASTA file;
  - Performs a multiple sequence alignment;
    - soaplab-based operations (seqret, emma, plot)
  - Renders the results



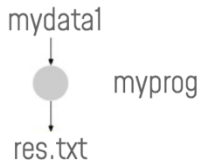
Historical Context

- The concept of workflows in science has evolved
  - from manual paper-based processes → highly automated systems
- This evolution has been driven by
  - increasing complexity of scientific research
  - processing & analysing large volumes of data.
- Scientific workflows are integral to a wide range of disciplines:
  - BioInformatics
  - Astronomy
  - NeuroInformatics
  - GeoSciences
  - Social Sciences ....

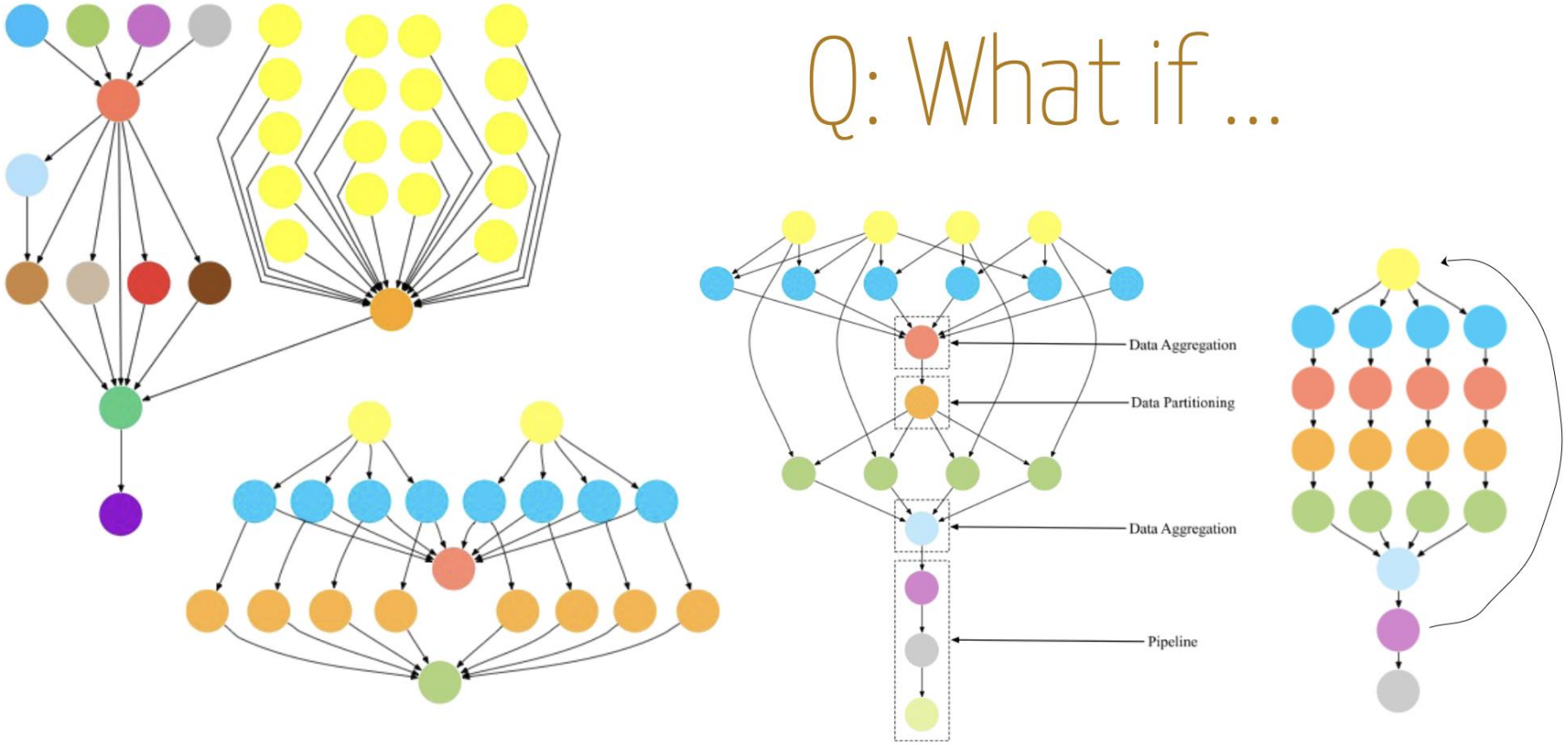
Historical Context

- What do you do when to want to run computations on a cluster ?
  - you write a submission script and submit it to the scheduler (e.g. Slurm) ?

```
#!/bin/bash
# Submission script for demonstrating
# slurm usage.

# Job parameters
#SBATCH --job-name=demo
#SBATCH --output=res.txt
# Needed resources
#SBATCH --ntasks=1
#SBATCH --mem-per-cpu=2000
#SBATCH --time=1:00:00

# Operations
echo "Job start at $(date)"
# Job steps
srun ~/bin/myprog < mydata1

echo "Job end at $(date)"
~
                          19,0-1        All
```



Historical Context



You need workflows !

You might use workflows already

Example of a Shell Script "*Workflow*"

```
#!/bin/bash
```

1) Stage-in input data to compute environment

```
scp myself@datastore.com:/data/input.txt /scratch/input.txt
```

2) Run a serial job with an input and output

```
bin/pre-processing in=input.txt out=tmp.txt
```

3) Run a parallel job with the resulting data

```
mpiexec bin/parallel-job in=tmp.txt out_prefix=output
```

4) Run a set of independent serial jobs in parallel – scheduling by hand

```
for i in `seq 0 $np`; do
      bin/integrity-check output.$i &
done
```

5) While those are running, get metadata and run another serial job

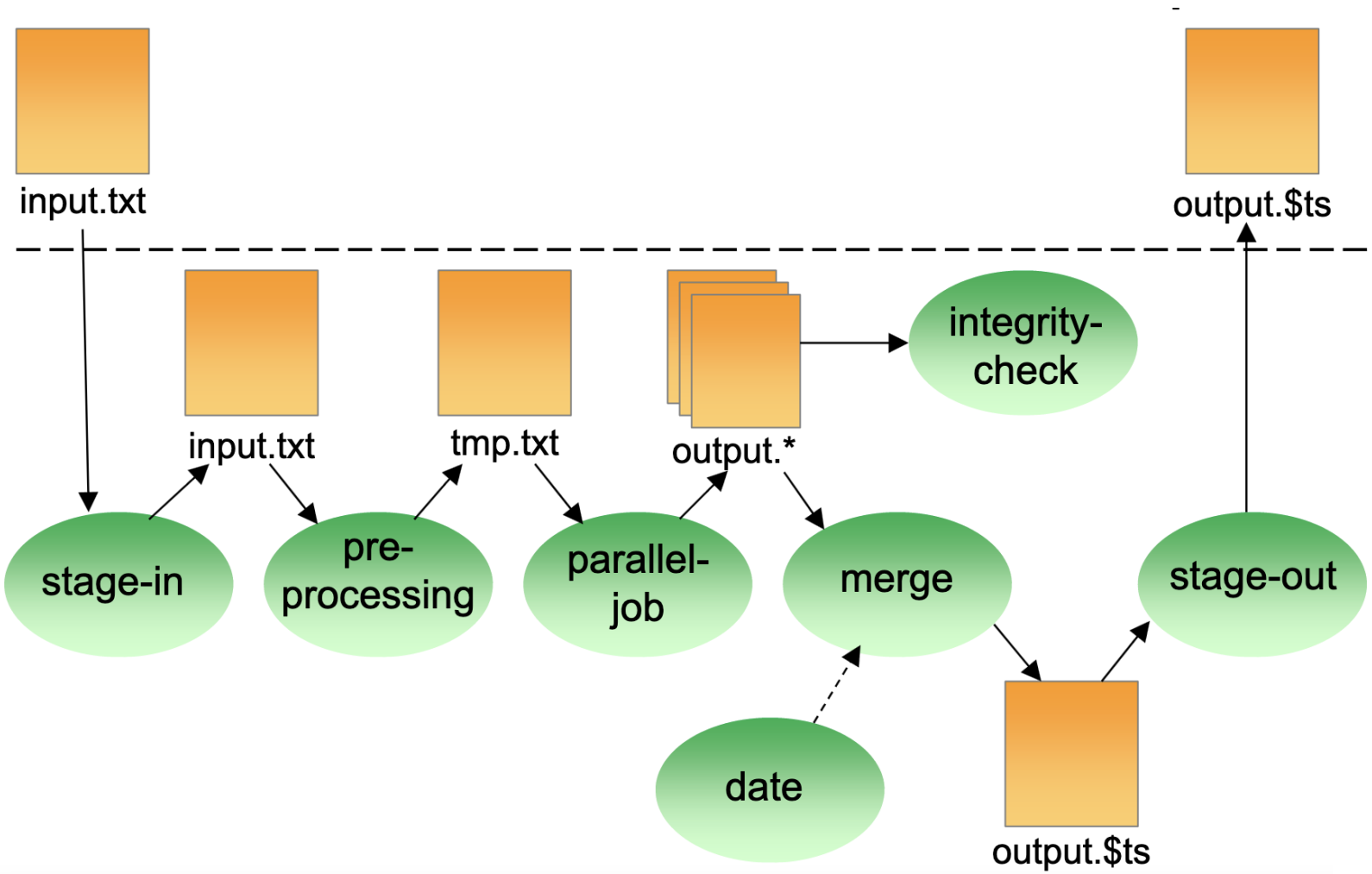
```
ts=`date +%s`
bin/merge prefix=output out=output.$ts
```

6) Finally, stage results back to permanent storage

```
scp /scratch/output.$ts myself@datastore.com:/data/output.$ts
```

You might use workflows already

Schematic of the Shell Script "*Workflow*"



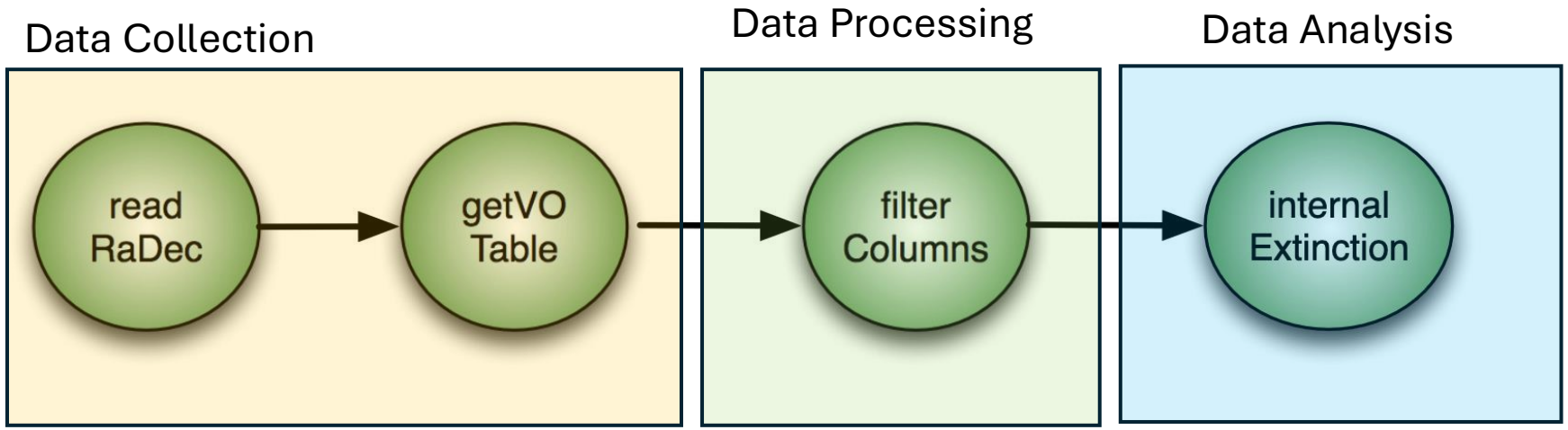
Workflows Key Phases

- Data Collection
  - acquiring raw data from experiments or simulations
- Data Processing
  - cleaning and structuring data for analysis
- Data Analysis
  - applying statistical models and algorithms to interpret data
- Visualization & Storage
  - creating graphical representations of analysis results
  - storing the results/metadata/provenance for further analysis

These components are interconnected in a workflow. Lets see an example!

Workflows Key Phases

- Astrophysics:
  - Calculating the Internal Extinction of Galaxies from the AMIGA catalogue
  - Crucial for determining a galaxy's optical luminosity.



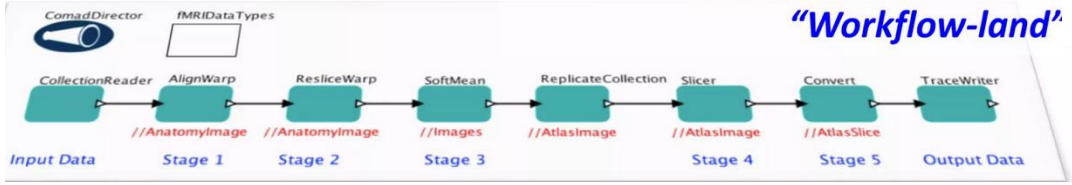
Workflow High-level Elements

- Task executions with dependencies
  - Specify a series of tasks to run
    - Data Collections, Data pre-processing and simulation
    - Data Analysis, Visualization, Storage
  - Outputs from one task may be inputs for another
    - DataFlow management – using streams/data between tasks
    - File management - using files between tasks
- Task scheduling
  - Some tasks may be able to run in parallel with other tasks
- Resource provisioning (getting processors)
  - Computational resources are needed to run jobs
- Ensuring accuracy and efficiency
  - Provenance generation – what is provenance in workflows ?

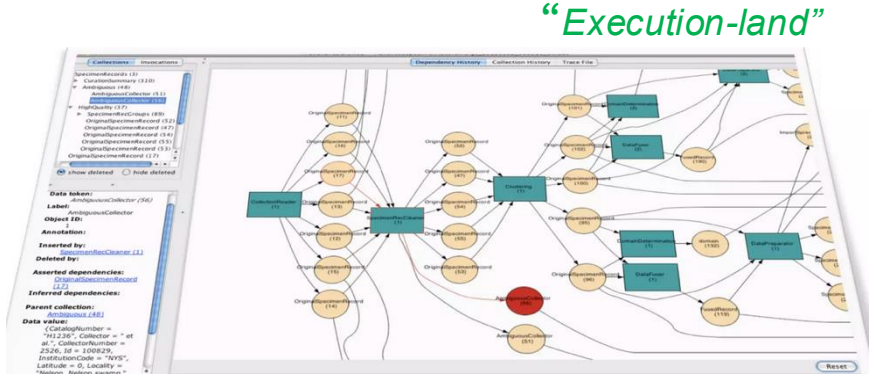
Workflow High-Level Elements

- Provenance:
  - Capturing information about the workflow
    - E.g. when a task was run ?; with which parameters?; what was the previous task ? And the next one ? ….
  - Enables reproducibility of results, transparency, data sharing
- Two types:
  - Prospective provenance
  - Retrospective provenance

- Prospective provenance:
  - captures the **structure and static context** of a workflow
  - it is independent of workflow execution or input data
  - provides an **abstract overview** of a workflow



- Retrospective provenance (~ tracing/profiling in HPC) :
  - captures the **information about a workflow's execution**
  - becomes available when **running the workflow**.
  - facts about the execution of each workflow step & runtime environment



What do we need help with?

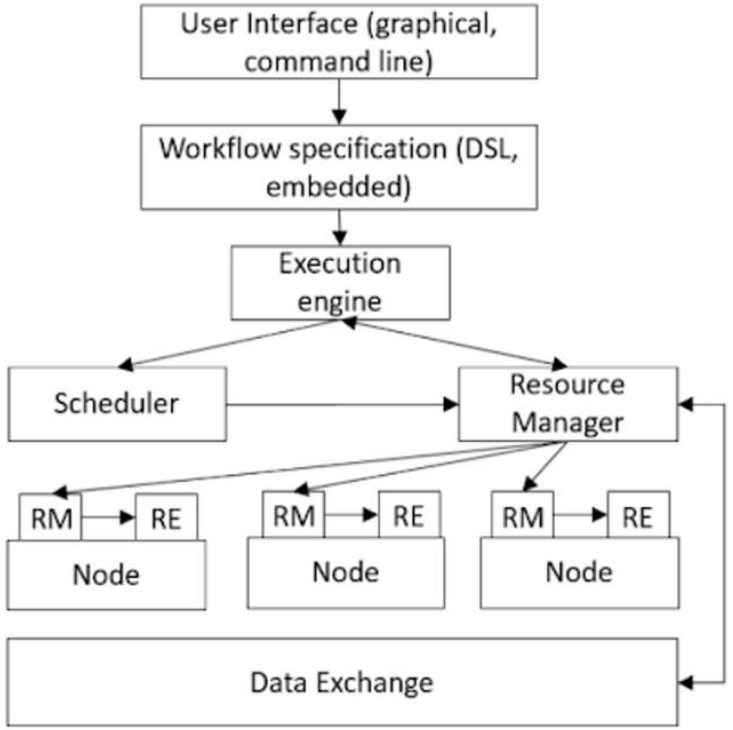
- Task executions with dependencies
  - What if something fails in the middle?
  - Dependencies may be complex
- Task scheduling
  - Minimize execution time while preserving dependencies
  - May have many tasks to run
- Resource provisioning
  - May want to run across multiple systems
  - How to match processors to work?

What do we need help with?

- Metadata and provenance
  - Automatically capture and track
    - Where did my task run? How long did it take?
    - What were the inputs and parameters?
    - What versions of code were used?
- File & Data management
  - Make sure inputs are available for tasks
  - Archive output data
- Automation
  - You have a workflow already – are there manual steps?

Workflow Management Systems (WMS)

- Workflow Management Systems (WMS)
  - Software solutions that provide the infrastructure for
    - Designing
    - Executing
    - Monitoring scientific workflows
  - They offer tools for
    - automating tasks
    - managing data flow
    - ensuring that computational resources are used efficiently
- Pegasus, Taverna, CWL, dispel4py, Apache Airflow, Nextflow, pycompss, ...
  - https://github.com/common-workflow-language/common-workflow-language/wiki/Existing-Workflow-systems
  - https://workflows.community/systems
  - Bringing the Scientific Workflows Community Together



Main components of a typical distributed infrastructure steered by a WMS

Workflow Management Systems (WMS)

https://github.com/common-workflow-language/common-workflow-language/wiki/Existing-Workflow-systems

common-workflow-language / common-workflow-language

Code | Issues 221 | Pull requests 20 | Actions | Wiki | Security | Insights

Existing Workflow systems

Stian Soiland-Reyes edited this page 3 weeks ago · 59 revisions

Permalink: https://s.apache.org/existing-workflow-systems

Cite as (update dates):

Peter Amstutz, Maxim Mikheev, Michael R. Crusoe, Nebojša Tijanić, Samuel Lampa, et al. (2024): **Existing Workflow systems.** *Common Workflow Language wiki*, GitHub. https://s.apache.org/existing-workflow-systems updated 2024-01-25, accessed 2024-01-25.

Computational Data Analysis Workflow Systems

An incomplete list

Please add new entries at the bottom.

In addition to this list, actively developed free/open-source systems should be registered at https://wo
systems

See also: https://github.com/pditommaso/awesome-pipeline

1. Arvados - CWL-based distributed computing platform for data analysis on massive data sets. http
https://github.com/arvados/arvados
2. Apache Taverna http://www.taverna.org.uk/ https://taverna.incubator.apache.org/
3. Galaxy http://galaxyproject.org/
4. SHIWA https://www.shiwa-workflow.eu/
5. Apache Oozie https://oozie.apache.org/
6. DNANexus https://wiki.dnanexus.com/API-Specification-v1.0.0/IO-and-Run-Specifications https://
API-Specification-v1.0.0/Workflows-and-Analyses

348 entries!

https://github.com/common-workflow-language/common-workflow-language/wiki/Existing-Workflow-systems 90%

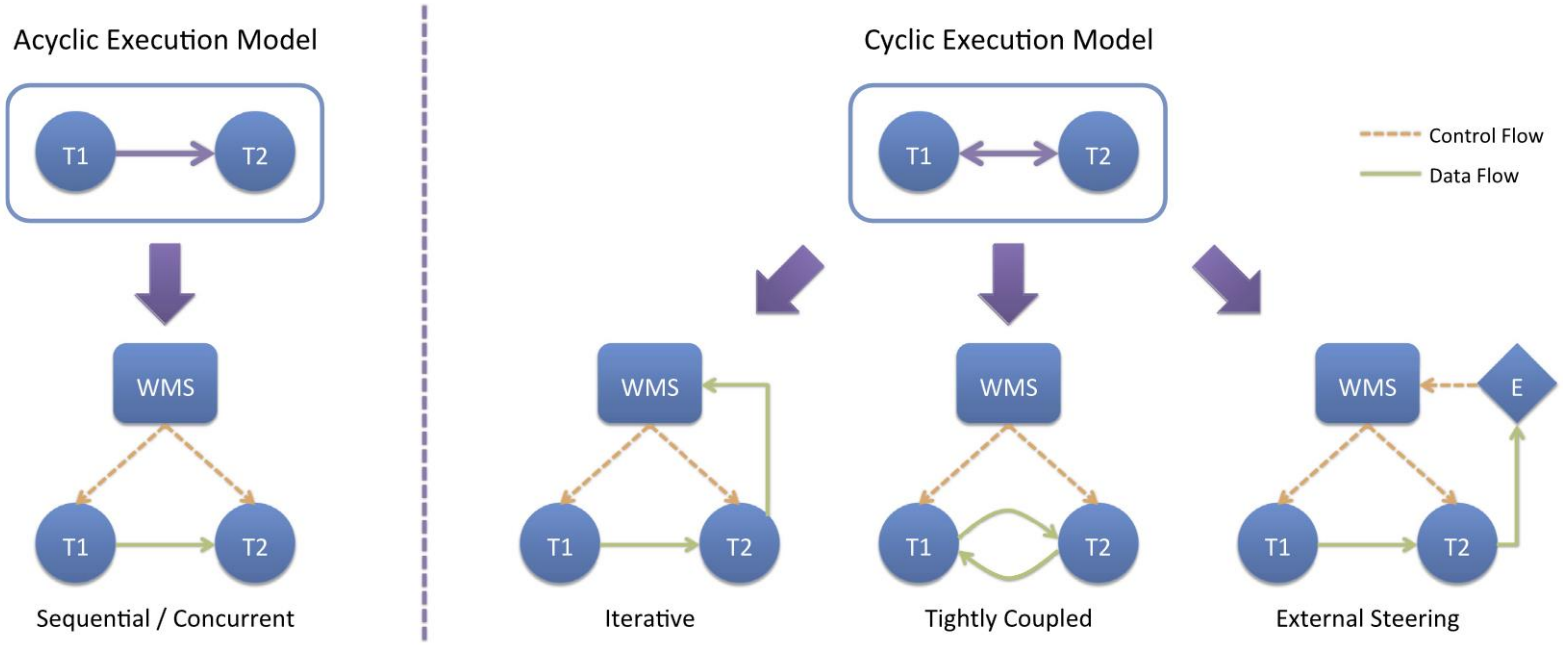
hpcflow

333. Data Version Control: DVC data pipelines as "Makefile" system for machine learning projects https://dvc.org/doc/start/data-management/data-pipelines
334. ZnFlow: General purpose framework for computational graphs https://github.com/zincware/ZnFlow
335. Globus Flows: Globus automation services provide secure and reliable task orchestration across a set of heterogeneous resources at scale. https://docs.globus.org/api/flows/
336. Kestra: Kestra is an infinitely scalable orchestration and scheduling platform, creating, running, scheduling, and monitoring millions of complex pipelines. https://github.com/kestra-io/kestra
337. Celery: Distributed Task Queue. https://github.com/celery/celery
338. dflow: Dflow is a Python framework for constructing scientific computing workflows (e.g. concurrent learning workflows) employing Argo Workflows as the workflow engine. https://github.com/deepmodeling/dflow
339. pydra: A simple dataflow engine with scalable semantics. https://github.com/nipype/pydra
340. funsies: A python library and execution engine to build reproducible, fault-tolerant, distributed and composable computational workflows. https://github.com/aspuru-guzik-group/funsies
341. Maize: A graph-based workflow manager for computational chemistry pipelines. https://github.com/MolecularAI/maize
342. doit: task management & automation tool. https://github.com/pydoit/doit
343. drama: asynchronous workflow executor engine based on Dramatiq and Apache Kafka https://github.com/KhaosResearch/drama
344. Titan: Semantic scientific workflow platform based on Drama, Apache Kafka, Avro and Spark https://titan.khaos.uma.es/
345. disBatch: Tool to distribute a list of computational tasks over a pool of compute resources. The pool can grow or shrink. https://github.com/flatironinstitute/disBatch
346. dawgz: Directed Acyclic Workflow Graph Scheduling. https://github.com/francois-rozet/dawgz
347. hummingqueue: Single tenant distributed scheduling. https://github.com/NablaChem/hummingqueue
348. Ewoks: Python-based meta workflow system for large-scale facilities, with multiple execution bindings https://ewoks.esrf.fr/ https://ewoks.readthedocs.io/

https://github.com/common-workflow-language/common-workflow-language/wiki/Existing-Workflow-systems

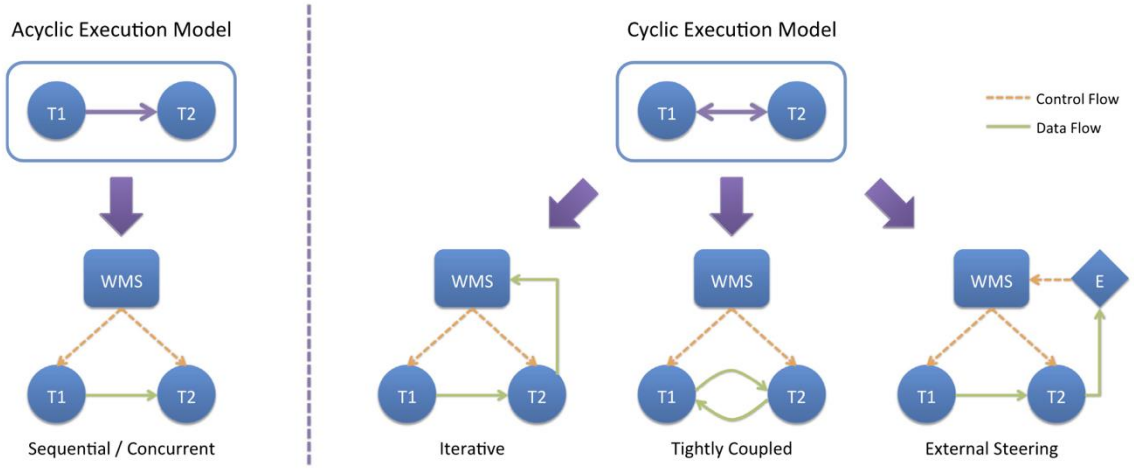
Workflow Management Systems (WMS)

Workflow execution model



A characterization of workflow management systems for extreme-scale applications

Workflow Management Systems (WMS)



- **Sequential**:
  - Begins with T1 followed by T2 upon completion;
  - Ideal for processes involving a simulation followed by post-processing.
- **Concurrent**:
  - T1 and T2 run simultaneously . T1's output feeding into T2;
  - Suitable for streaming data and multiple-stage processing
- **Iterative**:
  - Repeating sequences where T1's completion leads to T2, and T2's results may trigger further iterations
  - Useful for exploring parameter spaces through Uncertainty Quantification (UQ).
- **Tightly Coupled**:
  - T1 and T2 exchange partial results;
  - Supporting closely integrated simulations (e.g. multi-physics simulations that require tight coordination)
- **External Steering**:
  - Similar to concurrent but includes user/system evaluation of T2's output to potentially adjust T1 or other workflow aspects;
  - Useful for scenarios requiring real-time analysis feedback to guide subsequent steps

1.vi) Workflow Management Systems (WMS)

Classification of workflow management systems

Workflow Properties	ADIOS	Airavata	Askalon	Bobolang	dispel4py	Fireworks	Galaxy	Kepler	Makeflow	Moteur	Nextflow	Pegasus	Swift	Taverna	Triana
Workflow Execution Models															
Sequential	✔	✔	✔	✘	✘	✔	✔	✔	✔	✔	✘	✔	✔	✔	✔
Concurrent	✘	✘	✘	✔	✔	✘	✘	✘	✘	✘	✔	✘	✔	✘	✘
Iterative	✘	✘	✘	✘	✘	✘	✘	✘	✘	✔	✔	✘	✔	✔	✘
Tightly coupled	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
External steering	✘	✘	✘	✘	✘	✘	✘	✘	✘	✔	✘	✘	✘	✘	✘
Heterogeneous Computing Environments															
Co-location	✘	✘	✘	✔	✔	✔	✘	✘	✘	✘	✔	✔	✔	✘	✘
External location	✘	✔	✔	✘	✘	✘	✔	✔	✔	✔	✘	✔	✔	✔	✔
In situ	✔	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘	✘
Data Access Methods															
Memory	✔	✘	✘	✔	✔	✔	✘	✘	✘	✘	✔	✔	✔	✘	✘
Messages	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Local disk	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Shared file system	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
Object store	✘	✘	✘	✘	✘	✘	✔	✘	✘	✘	✘	✔	✘	✔	✘
Other remote storage	✔	✘	✘	✘	✘	✘	✔	✘	✘	✘	✘	✔	✔	✔	✘

A characterization of workflow management systems for extreme-scale applications

2. How to select a WMS ?

Key Factors to Consider

- Several WMS solve same general problems
  - but differ in specific approach
- A few categories to think :
  - Interface: how are workflows constructed?
  - Workload: what does your workflow look like?
  - Community: what domains does the tool focus on?
  - Task-Flow vs Data-Flow;

Interface

- How does a user construct workflows?
  - Graphical: like assembling a flow chart
  - Scripting: use a workflow tool-specific scripting language to describe workflow
  - API: use a common programming language with a tool-provided API to describe workflow
- Which is best depends on your application
  - Graphical can be unwieldy with many tasks
  - Scripting and API can require more initial investment
  - Some WMS support multiple approaches

Workload

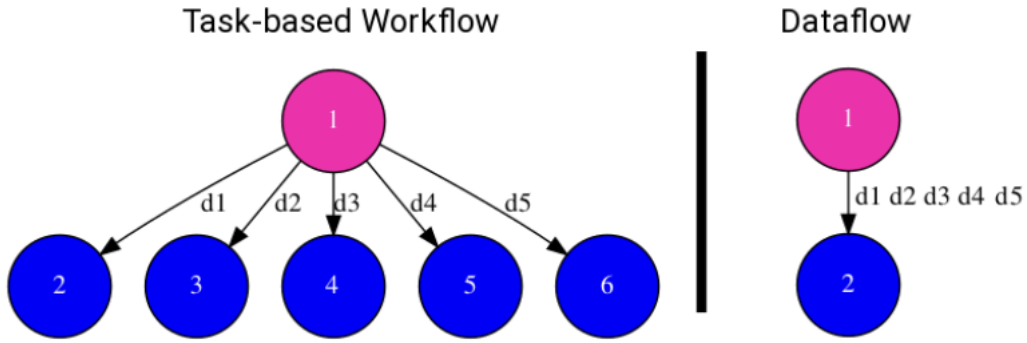
- What kind of workflow are you running?
  - Many vs. few tasks
  - Short vs. long
  - Dynamic vs. static
  - Loops vs. directed acyclic graph
- Different tools are targeted at different workload

Community

- What kinds of applications is the tool designed for?
- Some tools focus on certain science fields
  - Have specific paradigms or task types built-in
  - Workflow community will share science field
  - Less useful if not in the field or users of the provided tasks
- Some tools are more general
  - Open-ended, flexible
  - Less domain-specific community

Task-Flow vs Data-Flow

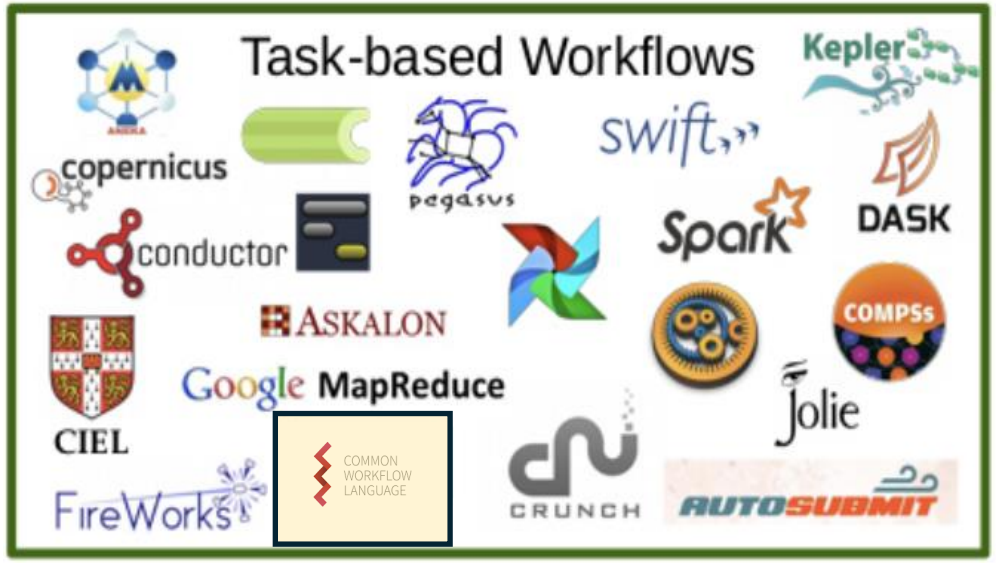
- Task-Flow:
  - Orchestrates isolated tasks processing data in sequence, based on data dependencies.
  - Ideal for structured, sequential processing where each task must complete before the next starts.
- Data-Flow:
  - Manages tasks for continuous data receipt and production;
  - Enables real-time data processing and inter-task communication without waiting for task completions.



- Key Difference:
  - Task-Flow focuses on sequential, dependent task execution, while Data-Flow emphasizes ongoing, real-time interaction between tasks

Task-Flow vs Data-Flow

dispel4py

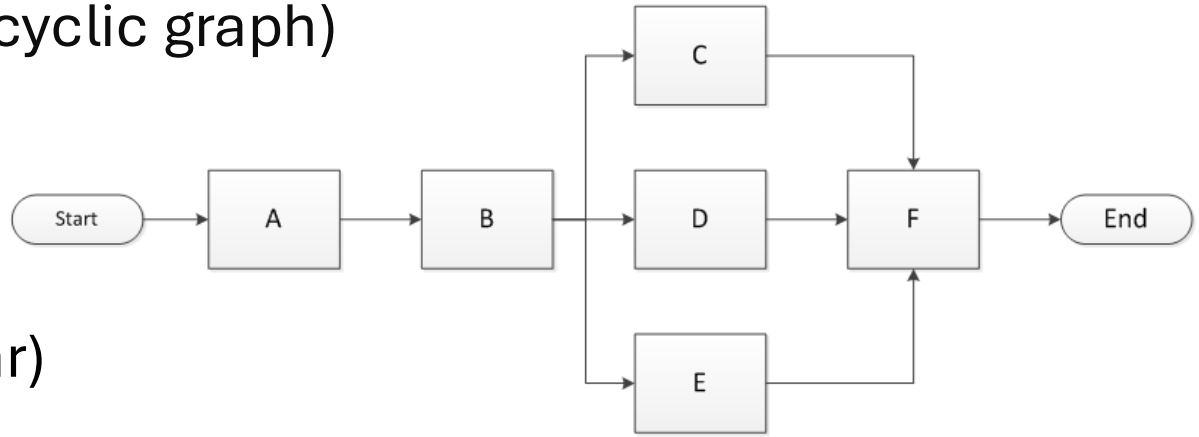


Common Workflow Language

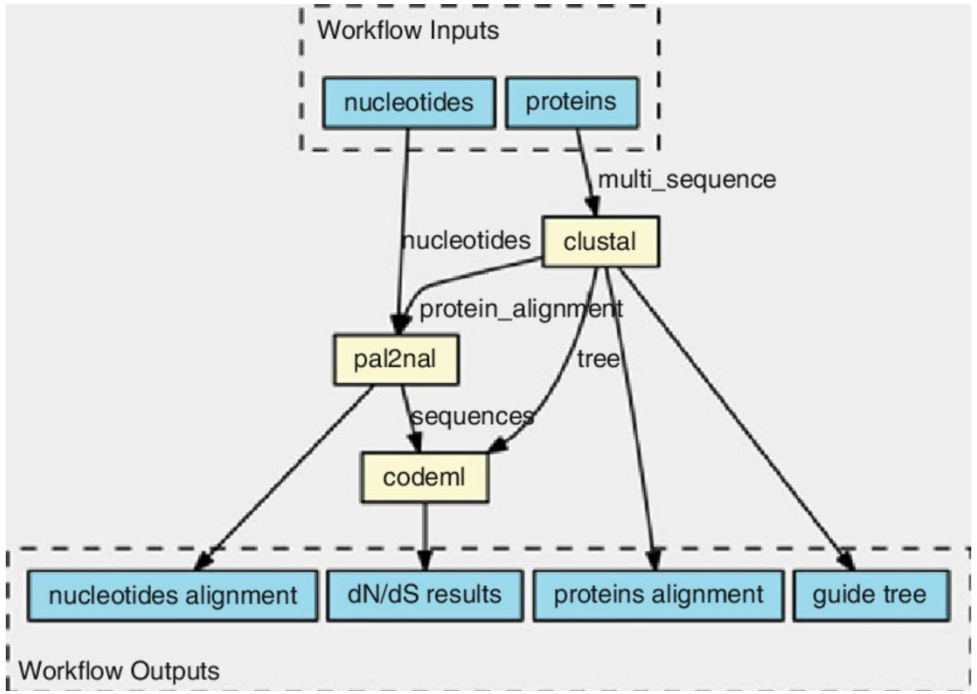
3. Workflows Concepts

Understanding DAG in Workflows

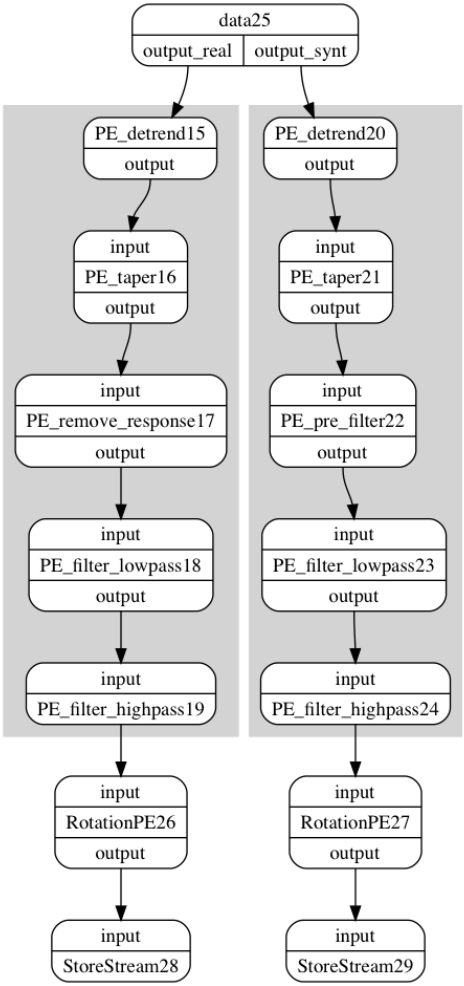
- Most WMS are based on DAG (directed acyclic graph)
  - vertices denote tasks
  - edges denote dependencies
  - unidirectional
  - never form a closed loop (non-circular)
- This non-circular structure ensures:
  - that each task is executed once
  - its prerequisites are met
  - efficient parallelization and scheduling



Understanding DAG in Workflows



Genomics CWL DAG workflow



Seismology dispel4py DAG workflow

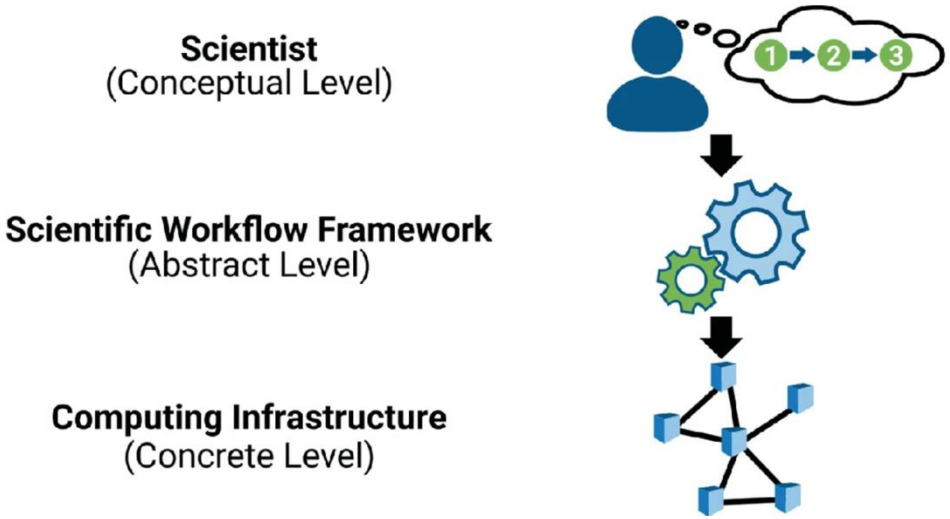
Levels of Composition

- Script-Gluing WMS:
  - **Description:** These workflows act as a 'glue' to connect standalone scripts/programs into a cohesive sequence.
  - **Usage:** Ideal for chaining existing programs or scripts that perform specific tasks into a workflow without modifying the original code.
  - **Examples:** Pegasus, CWL, Galaxy

- Programming Language-Like WMS:
  - **Description**: These workflows offer a programming language environment
    - Based on Python, C, Java, etc.
  - **Usage**: Suitable for workflows requiring complex decision-making, dynamic task generation, or intricate processing logic akin to software development.
  - **Examples**: dispel4py, PyCOMPSs, Dask, Nextflow

Levels of Workflow Abstraction

- **Conceptual Level:**
  - Captures the essence of the workflow's intent and the knowledge creation process.
  - Ideal for documenting and communicating the scientific approach in a structured manner.
- **Abstract Level**:
  - Defines the workflow structure and logic without binding to specific computational resources or environments.
  - Facilitates the interface between the scientific method and diverse computing infrastructures.
- **Concrete Level**:
  - Specifies exact details for execution, including data paths, environmental parameters, and computational resources.
  - Ensures workflows are ready to run in specific settings and are crucial for exploiting a wide range of data and computational platforms



Data Analysis and Exploration with Scientific Workflows

Computing vs. Semantic Focus

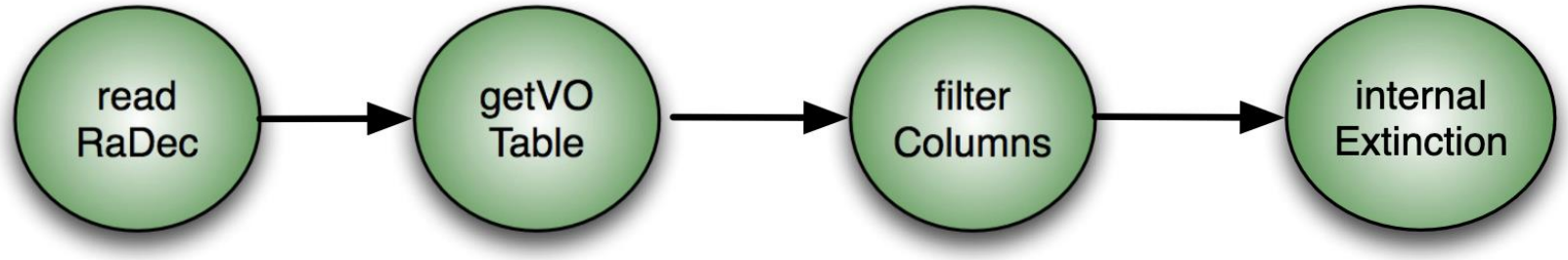
- Computing-centric workflows:
  - **Resource Optimization**: Streamlines computational resource use to
    - enhance performance
    - reduce costs
    - or both
  - **Execution Efficiency**: Focuses on the speed and reliability of tasks execution
    - incorporating strategies for error recovery and load balancing
  - **Examples**: dispel4py, Pegasus, Taverna

Computing vs. Semantic Focus

- Semantic-Focused Workflows:
  - **Data Meaning and Context**: Emphasizes the understanding of data by incorporating domain knowledge
    - often using ontologies to enrich data meaning.
  - **Data Integration and Interoperability**: Aims to seamlessly combine data from diverse sources and formats
    - facilitating collaboration across different scientific domains.
  - **Examples**: CWL, Wings
    - formal descriptions of data and operations
    - workflows are understandable by both machines and humans
    - supporting collaborative research efforts.

Real Examples – Internal Extinction (IE)

- Calculating the Internal Extinction of Galaxies
  - Crucial for determining a galaxy's optical luminosity
  - Build the full workflow in dispel4py
  - Each task/step in IE follows a Data-Flow pattern



dispel4py: A Python framework for data-intensive scientific computing

Acknowledgements

- *Overview of Scientific Workflows, Why Use them?*, Scott Callaghan
- *Introduction to Scientific Workflows and Pegasus*, Karan Vahi
- *Workflows and Provenance: Toward Information Science Solutions for the Natural Sciences* , B. Ludaescher
- *DARE: A Reflective Platform Designed to Enable Agile Data-Driven Research on the Cloud*, DARE Team
- *Workflows Community Summits* , Workflows Community
- *A characterization of workflow management systems for extreme-scale applications*